Mobile Application Development

Practical No. 15

Design and develop a simple countdown timer.

Que. 1 - Create a countdown timer that starts from 10 seconds and updates every second.

Que. 2. Display the remaining time in TextView and show "Time's up!" when finished.

Que. 3. Add a reset button to restart the timer from the beginning.

Que. 4. Modify the timer to accept user input for duration (e.g. from EditText).

Que. 5. Implement a timer that disables a button until countdown completes.

Que. 6. Extend the timer to trigger a sound or vibration when finished.

ANS:-

1 - activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:id="@+id/tvTimer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="10"
        android:textSize="40sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/etSeconds"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter seconds"
        android:inputType="number"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnStart"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start Timer"
        android:layout_marginBottom="10dp"/>
```

```
    <Button
        android:id="@+id/btnReset"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Reset Timer"/>

</LinearLayout>
```

2 - MainActivity.java

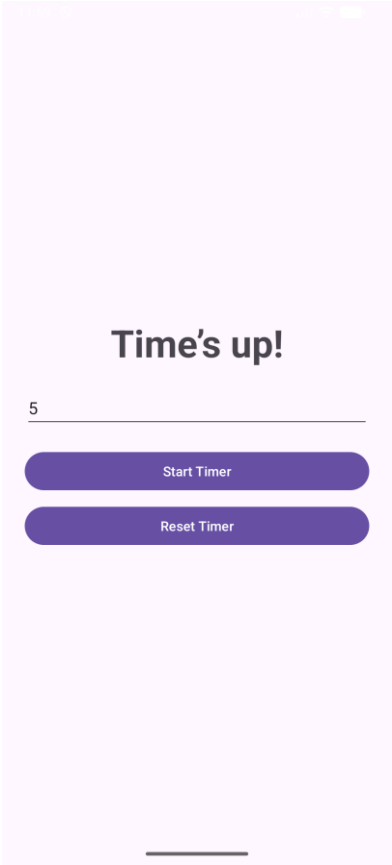
```
package com.example.practical15;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.os.CountDownTimer;
import android.os.VibrationEffect;
import android.os.Vibrator;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView tvTimer;
    EditText etSeconds;
    Button btnStart, btnReset;

    CountDownTimer countDownTimer;
    long startTime = 10000; // Default 10 seconds
    long timeLeft = startTime;
    boolean timerRunning = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvTimer = findViewById(R.id.tvTimer);
        etSeconds = findViewById(R.id.etSeconds);
        btnStart = findViewById(R.id.btnStart);
        btnReset = findViewById(R.id.btnReset);

        updateTimerText();

        btnStart.setOnClickListener(v -> {

            if (timerRunning) return;

            String input = etSeconds.getText().toString().trim();

            if (!input.isEmpty()) {
                try {
                    startTime = Long.parseLong(input) * 1000;
                } catch (NumberFormatException e) {
                    etSeconds.setError("Invalid number");
                    return;
                }
            }

            if (startTime <= 0) {
                etSeconds.setError("Enter positive seconds");
                return;
            }
```

```
            timeLeft = startTime;
            startTimer();
        });

        btnReset.setOnClickListener(v -> resetTimer());
    }

    private void startTimer() {

        btnStart.setEnabled(false); // Disable button during countdown
        timerRunning = true;

        countDownTimer = new CountDownTimer(timeLeft, 1000) {
            @Override
            public void onTick(long millisUntilFinished) {
                timeLeft = millisUntilFinished;
                updateTimerText();
            }

            @Override
            public void onFinish() {
                timerRunning = false;
                tvTimer.setText("Time’s up!");
                btnStart.setEnabled(true);

                playSound();
                vibratePhone();
            }
        }.start();
    }

    private void resetTimer() {
        if (countDownTimer != null) {
            countDownTimer.cancel();
        }
        timerRunning = false;
        timeLeft = 10000; // Reset to 10 seconds default
        updateTimerText();
        btnStart.setEnabled(true);
    }

    private void updateTimerText() {
        int seconds = (int) (timeLeft / 1000);
        tvTimer.setText(String.valueOf(seconds));
    }

    private void playSound() {
        MediaPlayer mp = MediaPlayer.create(this, R.raw.alarm);
        mp.start();
    }

    private void vibratePhone() {
        Vibrator vibrator = (Vibrator) getSystemService(VIBRATOR_SERVICE);
        if (vibrator != null) {
            if (android.os.Build.VERSION.SDK_INT >=
android.os.Build.VERSION_CODES.O) {
                vibrator.vibrate(VibrationEffect.createOneShot(1000,
VibrationEffect.DEFAULT_AMPLITUDE));
            } else {
                vibrator.vibrate(1000);
            }
        }
    }
}
```

3 – AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.VIBRATE"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Practical15">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER"
/>
            </intent-filter>
        </activity>
    </application>

</manifest>
```

4 – Create raw File:-

File Path:- res/raw/alarm.mp3

Output:-